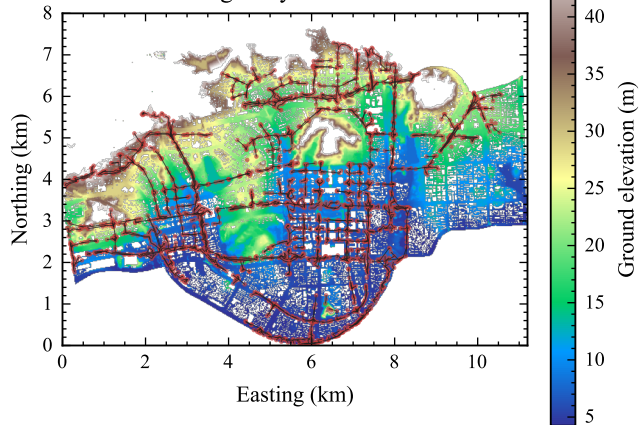
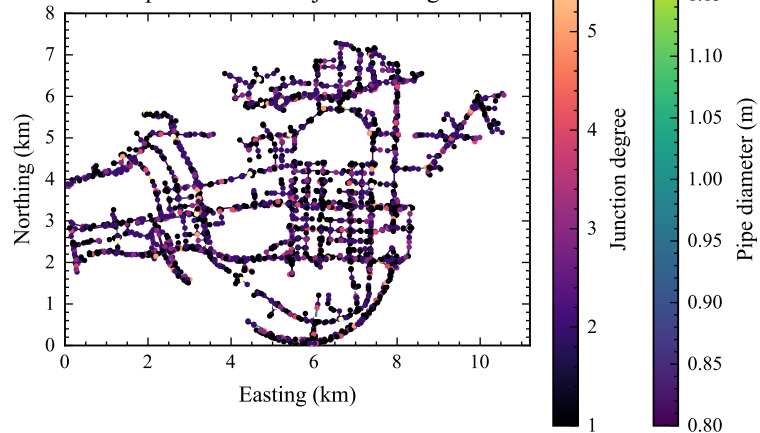


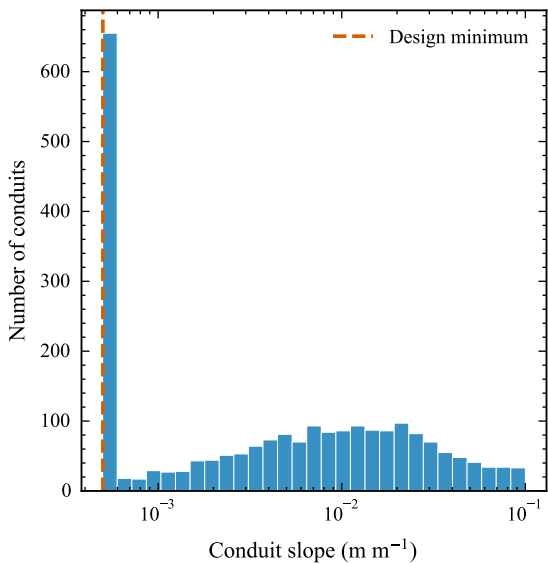
(a) Terrain and gravity-consistent network



(b) Pipe diameter and junction degree



(c) Positive-slope constraint



(d) Geometry of retained conduits

